

Assignment

Value **100%** of Coursework Resit

Individual work

Background

The subway system of a city is a network of underground or elevated trains that provide rapid transit for passengers within the urban area. It typically consists of interconnected lines or routes, each with designated stops or stations where passengers can board or disembark. Subway systems are a vital component of urban transportation infrastructure, offering a convenient and reliable mode of travel for commuters, tourists, and residents alike. Subway stations are strategically located at key points throughout the city.

There are several terms used to describe the subway system:

- **Station:** is a designated node along a subway line where trains stop to allow passengers to board or alight.
- **Terminal Station:** is the endpoint of a subway line or route where trains begin or end their journeys.
- **Interchange Station:** is a crucial component of a subway system. It is a station where passengers can transfer between different subway lines or routes without exiting the transit system.
- **Route:** refers to the path that a subway train follows between two or more stations.
- **Line:** is a set of interconnected routes that share a common designation and usually serve a specific geographic area or corridor within a city. Each line is identified by a unique name or number and is represented by a distinct color on maps and signage.

General requirements

This coursework is to write a project to simulate a subway system according to one subway map provided as an appendix file named *Subway Map of Chengdu(2019).png*. The map depicts the operation of metro lines in Chengdu as of the end of 2019. It provides the information the project needs.

The following figure is a thumbnail of the map. You'd better view the appendix file directly for the high-resolution details.



Note:

You must create the corresponding graph based on the appendix file exactly.

- Node

You only need to consider the termination station and interchange station in the map as nodes of the graph.

- Route

You only need to consider the colored lines in the map as a route of the graph, please ignore the grey lines that mean the planning route not in service at that time.

- Edge

The distance/cost between two nodes should be labeled as the number of sections that exist in real.

Submission Format Requirements

When you have completed all tasks, you should be able to generate one executable project, otherwise you will lose all marks.

The coursework submitted should be composed of two files:

- a *report* as a *Microsoft Word* document containing a description and explanation of the encoding work.
 - filename format: *Student ID_CHC5223_CW_Resit_Report.docx*
- a *.zip file* containing the project source code files:
 - all the project's source files, including those provided.
 - filename format: *Student ID_CHC5223_CW_Resit_Project.zip*

If you do not submit the files according to the requirements, you may lose 10 marks for the coursework.

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Requirements

Task 1

Please fill in the spreadsheet of the Excel file *adjacency matrix.xlsx* provided to complete the adjacency matrix according to the map.

Each node should be named as the abbreviation of the first letters of the station name shown on the map. For instance, the station *Weijianian* should be instantiated as *wjn*.

Each edge should be labeled as the distance(cost) between two nodes. For instance, the edge between Weijianian Station and North Railway Station should be labeled as 2. If there is not an edge between two nodes or on itself, distance/cost is "Infinite".

Tip: There are several examples written for reference in the adjacency matrix.

10 marks

Task 2

Please create one project in the IDE based on the start code files provided. You may modify the files to make the project executable upon completion.

5 marks

Task 3

The class **Station** is a class defined to represent subway stations that are the terminal stations or interchange stations. The attribution 'name' represents the name of the subway station. The attribution 'line' is an array list to record the line numbers that the station is located at. The attribution 'previous' is used to record the previous station in which the shortest path. The attribution 'g_value' is used to record the g_value for the pathfinding algorithm. The attribution 'previous' is used to record which node is the previous one during finding the shortest path. The attribution 'isTermination' is used to represent the station as a termination station(true) or interchange station(false).

You must write the corresponding code in the Station.java file based on the notations.

10 marks

Explain the encoding work in the report rationally and explicitly.

10 marks

Task 4

The class **subwayMap** is used to generate the graph that represents the subway map. The nodes of the graph are the termination stations and interchange

stations of the subway. The edges of the graph are the lines between two stations. The attribution 'stations' is used to record all stations that exist on the subway map. The attribution 'stationIndex' represents the index of one station. The attribution 'stationMap' is used to record the pairing key-value of the station and station index. The attribution 'size' represents the number of stations. The 'adjacencyMatrix' is a two-dimensional array to record all edges' distance(cost). You must write the corresponding codes in the *subwayMap.java* file based on the notations.

10 marks

Explain the encoding work in the report rationally and explicitly.

10 marks

Task 5

The class **Main** is used to generate a graph that can represent the subway of Chengdu at the end of 2019 according to the appendix file.

- In this class, you need to create all stations according to the appendix file. In this class, you need to create an object of subwayMap to represent the subway system of Chengdu according to the appendix file.

5 marks

Task 6

In the **Main** class, you need to implement the DFT(depth-first traversal) and BFT(breadth-first traversal) algorithms to traverse all stations on the map.

- The implementations of DFT and BFT algorithms can display the process of the traversal, including station name, station type, and subway line number information in the sequence of the traversal.
- The station type should be displayed as 'termination' or 'interchange' according to the station's status. There is an example shown for reference in the following screenshot.

```

-----DepthFirstTraversal information-----
(station name)          (station type)          (line number)
Weijianian              termination              1
North Railway           interchange              1 7
Luomashi                interchange              1 4
-----BreadthFirstTraversal information-----
(station name)          (station type)          (line number)
Weijianian              termination              1
North Railway           interchange              1 7
Luomashi                interchange              1 4

```

- The start station must be the 'Tianfu Square' station.

10 marks

Explain the encoding work in the report rationally and explicitly.

10 marks

Task 7

In the **Main** class, you need to implement Dijkstra's algorithm to find the shortest path from 'East Chengdu Railway Station' to 'Taipingyuan' station.

- The implementations of Dijkstra's algorithms can display the information of the pathfinding process, including the station name of each station in the closed set and open set, the g_value of the station node, or other information that may be necessary. There is an example shown for reference in the following screenshot.

```
open has
North Railway    g value is 2
open has
Luomashi         g value is 5
open has
closed has
North Railway    g value is 2
Luomashi         g value is 5
Weijianian       g value is 0
```

- The implementation of Dijkstra's algorithm can display the shortest path found. There is an example shown for reference in the following screenshot.

```
The shortest path from Weijianian to Luomashi is
Weijianian -> North Railway -> Luomashi
```

Tip: You may modify the **Station** class to make each object of it can be comparable.

10 marks

Explain the encoding work in the report rationally and explicitly.

10 marks